

1. (9 pts) For all a and b in $\mathbb{Z}$, define $a \sim b$ if and only if $a-b$ is a multiple of 4. Prove that $\sim$ is an equivalence relation on $\mathbb{Z}$.

$\sim$ is reflexive: If $a \in \mathbb{Z}$ we know $a-a=0$ is a multiple of 4 so $a \sim a$.

$\sim$ is symmetric: For any a, b in $\mathbb{Z}$ if $a \sim b$ then $b \sim a$.

Let $a \sim b$. This means $a-b=4k$ for some $k \in \mathbb{Z}$.

Therefore $b-a = -(a-b) = -4k = 4(-k)$ is a multiple of 4

since $-k \in \mathbb{Z}$ when k is.

so $b \sim a$ whenever $a \sim b$.

$\sim$ is transitive: For any a, b, c in $\mathbb{Z}$, if $a \sim b$ and $b \sim c$ then $a \sim c$.

Let $a \sim b$ and $b \sim c$. This means $a-b=4j$ and $b-c=4k$ for some j and k in $\mathbb{Z}$.

Therefore $a-c = (a-b) + (b-c) = 4j + 4k = 4(j+k)$ is a multiple of 4 since $j+k \in \mathbb{Z}$ when both j & k are integers.

so $a \sim c$ whenever $a \sim b$ and $b \sim c$

this is an equivalence relation since it satisfies all three properties.

2. (3 pts) Describe 3 different equivalence classes for the relation in #1.

There are 4 different equivalence classes for this relation.

$[0] =$ all integers x such that $x-0$ is a multiple of 4.

$$= \{4k : k \in \mathbb{Z}\}$$

$$[1] = \{4k+1 : k \in \mathbb{Z}\}$$

Over $\rightarrow$

$$[2] = \{4k+2 : k \in \mathbb{Z}\}$$

$$[3] = \{4k+3 : k \in \mathbb{Z}\}$$

3. (4 pts) What does it mean to say the equivalence classes of a relation defined on a set S will partition S ?

The equivalence classes partition S means every element of S is in exactly one class.

4. (4pts) For a and b in $\mathbb{Z}$, define the relation $a \sim b$ if and only if $a > b$. Which of the three equivalence relation properties **fail to hold** for this relation?

$\sim$ is not reflexive since a is not related to a .
 $a \not\sim a$ a is not greater than a .

$\sim$ is not symmetric. If $a \sim b$ we know $a > b$.
But then b cannot be greater than a . So b is not related to a .

$\sim$ is transitive since $a > b$ and $b > c$ implies $a > c$